

Session 5

Input Devices: Joystick, IR & Buttons



Turn a stick and a button into a menu you can steer.

Part	Topic	Time
01	The joystick: read it, then find your limits	8 min
02	Zones and edges: one push, one move	7 min
03	Practice: a joystick menu on the Serial Monitor	12 min
04	The IR receiver: what is coming	3 min

Builds on plotting the joystick in Session 2, arrays from Lesson 4, and switch and counting on release from M2 Session 2.

Five words you will hear today.

Word	Plain meaning
Raw value	The number exactly as the module gives it, before you decide what it means.
Limit	A number you pick: past it, the stick counts as pushed.
Zone	Which part of its travel the stick is in: pushed one way, resting, or pushed the other way.
Edge	The moment something changes — the stick just got pushed, the button just got let go.
Wrap around	Going past the last menu item brings you back to the first, like the hands of a clock.

A joystick is a game controller's thumbstick.

Push it and the game moves; let go and it springs back to the middle.

The board can't see "left" or "right" — it gets a number for how far the stick is pushed, and you decide what each number means.

Three readings, one call each.

ONE LINE OF CODE

```
float x = incipe.getJoystickXMovement();
float y = incipe.getJoystickYMovement();
float button = incipe.getJoystickButtonResponse();
```

Call	Value	Not detected
incipe.getJoystickXMovement()	A float, raw left–right position	-1
incipe.getJoystickYMovement()	A float, raw up–down position	-1
incipe.getJoystickButtonResponse()	A float, raw stick-button reading	-1

The range, the centre and which button value means “pressed” are not fixed — find them in the Serial Plotter. Leave the next connection free: the joystick uses it for Y and the button. Wiki › Joystick.

Find your limits.

Step	Do this
1	Plot X in the Serial Plotter, as in Session 2.
2	Write down three numbers: X at rest, X pushed fully one way, X pushed fully the other way.
3	Lower limit: halfway between rest and the smaller end. Upper limit: halfway between rest and the bigger end.

Today's code uses **25** and **75** as stand-in limits so the examples are easy to follow. They are not your joystick's numbers — replace them with yours.

One push, one move — like a doorbell.

Holding a doorbell down rings it once, not a hundred times.

A menu should work the same way: one push of the stick moves one item, however long you hold it.

Sort a reading into a zone.

```
1  const float LOW_LIMIT = 25;    // change to your number
2  const float HIGH_LIMIT = 75;   // change to your number
3
4  int zone(float x) {
5      if (x == -1) return 0;      // not detected: treat as resting
6      if (x < LOW_LIMIT) return -1; // pushed one way
7      if (x > HIGH_LIMIT) return 1; // pushed the other way
8      return 0;                  // resting in the middle
9  }
```

LINE 1 • STEP 1 OF 6

Your lower limit. Below it, the stick is pushed one way.

VARIABLES

LOW_LIMIT = 25

Sort a reading into a zone.

```
1  const float LOW_LIMIT = 25;    // change to your number
2  const float HIGH_LIMIT = 75;   // change to your number
3
4  int zone(float x) {
5      if (x == -1) return 0;      // not detected: treat as resting
6      if (x < LOW_LIMIT) return -1; // pushed one way
7      if (x > HIGH_LIMIT) return 1; // pushed the other way
8      return 0;                  // resting in the middle
9  }
```

LINE 2 • STEP 2 OF 6

Your upper limit. Above it, pushed the other way.

VARIABLES

LOW_LIMIT = 25

HIGH_LIMIT = 75

Sort a reading into a zone.

```
1  const float LOW_LIMIT = 25;    // change to your number
2  const float HIGH_LIMIT = 75;   // change to your number
3
4  int zone(float x) {
5      if (x == -1) return 0;      // not detected: treat as resting
6      if (x < LOW_LIMIT) return -1; // pushed one way
7      if (x > HIGH_LIMIT) return 1; // pushed the other way
8      return 0;                  // resting in the middle
9  }
```

LINE 4 • STEP 3 OF 6

Call zone(90): x starts as 90.

VARIABLES

LOW_LIMIT = 25
HIGH_LIMIT = 75
x = 90

Sort a reading into a zone.

```
1  const float LOW_LIMIT = 25;    // change to your number
2  const float HIGH_LIMIT = 75;   // change to your number
3
4  int zone(float x) {
5      if (x == -1) return 0;       // not detected: treat as resting
6      if (x < LOW_LIMIT) return -1; // pushed one way
7      if (x > HIGH_LIMIT) return 1; // pushed the other way
8      return 0;                  // resting in the middle
9  }
```

LINE 5 • STEP 4 OF 6

Is x -1, “not detected”? No.

VARIABLES

```
LOW_LIMIT = 25
HIGH_LIMIT = 75
x = 90
```

Sort a reading into a zone.

```
1  const float LOW_LIMIT = 25;    // change to your number
2  const float HIGH_LIMIT = 75;   // change to your number
3
4  int zone(float x) {
5      if (x == -1) return 0;      // not detected: treat as resting
6      if (x < LOW_LIMIT) return -1; // pushed one way
7      if (x > HIGH_LIMIT) return 1; // pushed the other way
8      return 0;                  // resting in the middle
9  }
```

LINE 6 • STEP 5 OF 6

Is 90 below 25? No.

VARIABLES

```
LOW_LIMIT = 25
HIGH_LIMIT = 75
x = 90
```

Sort a reading into a zone.

```
1  const float LOW_LIMIT = 25;    // change to your number
2  const float HIGH_LIMIT = 75;   // change to your number
3
4  int zone(float x) {
5      if (x == -1) return 0;       // not detected: treat as resting
6      if (x < LOW_LIMIT) return -1; // pushed one way
7      if (x > HIGH_LIMIT) return 1; // pushed the other way
8      return 0;                  // resting in the middle
9  }
```

LINE 7 • STEP 6 OF 6

Is 90 above 75? Yes — hand back 1: pushed.

VARIABLES

```
LOW_LIMIT = 25
HIGH_LIMIT = 75
x = 90
result = 1
```

Act when the zone changes, not while it stays.

THE PROBLEM

While you hold the stick, every pass of `loop()` sees zone 1 again — the menu would race through every item.

THE EDGE

Act only when the zone is not 0 now, but was 0 last pass: the stick was just pushed.

THE TRICK

Keep last pass's zone in a variable, `lastZone`, and update it at the end of every pass.

Build a joystick menu on the Serial Monitor.

The curriculum task: a simple menu system controlled by the Joystick and displayed on the Serial Monitor.

The menu is an array of three words; choice is the box you are on.

The menu, one pass at a time.

```
1  const char* items[] = {"Temperature", "Humidity", "Light"};
2  int choice = 0;
3  int lastZone = 0;
4
5  void loop() {
6      int z = zone(incipe.getJoystickXMovement());
7      if (z != 0 && lastZone == 0) {    // just pushed
8          choice = choice + z;
9          if (choice > 2) choice = 0;      // past the end: back to the start
10         if (choice < 0) choice = 2;      // before the start: to the end
11         Serial.println(items[choice]);
12     }
13     lastZone = z;
14     delay(50);
15 }
```

LINE 1 • STEP 1 OF 19

A list of three menu items. Box 0 holds “Temperature”.

VARIABLES

SERIAL MONITOR

The menu, one pass at a time.

```
1  const char* items[] = {"Temperature", "Humidity", "Light"};
2  int choice = 0;
3  int lastZone = 0;
4
5  void loop() {
6      int z = zone(incipe.getJoystickXMovement());
7      if (z != 0 && lastZone == 0) {    // just pushed
8          choice = choice + z;
9          if (choice > 2) choice = 0;      // past the end: back to the start
10         if (choice < 0) choice = 2;      // before the start: to the end
11         Serial.println(items[choice]);
12     }
13     lastZone = z;
14     delay(50);
15 }
```

LINE 2 • STEP 2 OF 19

Start on item 0.

VARIABLES

choice = 0

SERIAL MONITOR

The menu, one pass at a time.

```
1  const char* items[] = {"Temperature", "Humidity", "Light"};
2  int choice = 0;
3  int lastZone = 0;
4
5  void loop() {
6      int z = zone(incipe.getJoystickXMovement());
7      if (z != 0 && lastZone == 0) {    // just pushed
8          choice = choice + z;
9          if (choice > 2) choice = 0;    // past the end: back to the start
10         if (choice < 0) choice = 2;    // before the start: to the end
11         Serial.println(items[choice]);
12     }
13     lastZone = z;
14     delay(50);
15 }
```

LINE 3 • STEP 3 OF 19

Last pass's zone: resting.

VARIABLES

choice = 0

lastZone = 0

SERIAL MONITOR

The menu, one pass at a time.

```
1  const char* items[] = {"Temperature", "Humidity", "Light"};
2  int choice = 0;
3  int lastZone = 0;
4
5  void loop() {
6      int z = zone(incipe.getJoystickXMovement());
7      if (z != 0 && lastZone == 0) {    // just pushed
8          choice = choice + z;
9          if (choice > 2) choice = 0;    // past the end: back to the start
10         if (choice < 0) choice = 2;    // before the start: to the end
11         Serial.println(items[choice]);
12     }
13     lastZone = z;
14     delay(50);
15 }
```

LINE 6 • STEP 4 OF 19

Pass 1: the stick rests, the reading is 50, so the zone is 0.

VARIABLES

```
choice = 0
lastZone = 0
z = 0
```

SERIAL MONITOR

The menu, one pass at a time.

```
1  const char* items[] = {"Temperature", "Humidity", "Light"};
2  int choice = 0;
3  int lastZone = 0;
4
5  void loop() {
6      int z = zone(incipe.getJoystickXMovement());
7      if (z != 0 && lastZone == 0) {    // just pushed
8          choice = choice + z;
9          if (choice > 2) choice = 0;    // past the end: back to the start
10         if (choice < 0) choice = 2;    // before the start: to the end
11         Serial.println(items[choice]);
12     }
13     lastZone = z;
14     delay(50);
15 }
```

LINE 7 • STEP 5 OF 19

Is z not 0? No — skip the block. Line 13 keeps lastZone at 0.

VARIABLES

```
choice = 0
lastZone = 0
z = 0
```

SERIAL MONITOR

The menu, one pass at a time.

```
1  const char* items[] = {"Temperature", "Humidity", "Light"};
2  int choice = 0;
3  int lastZone = 0;
4
5  void loop() {
6      int z = zone(incipe.getJoystickXMovement());
7      if (z != 0 && lastZone == 0) {    // just pushed
8          choice = choice + z;
9          if (choice > 2) choice = 0;    // past the end: back to the start
10         if (choice < 0) choice = 2;    // before the start: to the end
11         Serial.println(items[choice]);
12     }
13     lastZone = z;
14     delay(50);
15 }
```

LINE 6 • STEP 6 OF 19

Pass 2: you push. The reading is 90, so the zone is 1.

VARIABLES

choice = 0
lastZone = 0
z = 1

SERIAL MONITOR

The menu, one pass at a time.

```
1  const char* items[] = {"Temperature", "Humidity", "Light"};
2  int choice = 0;
3  int lastZone = 0;
4
5  void loop() {
6      int z = zone(incipe.getJoystickXMovement());
7      if (z != 0 && lastZone == 0) {    // just pushed
8          choice = choice + z;
9          if (choice > 2) choice = 0;      // past the end: back to the start
10         if (choice < 0) choice = 2;      // before the start: to the end
11         Serial.println(items[choice]);
12     }
13     lastZone = z;
14     delay(50);
15 }
```

LINE 7 • STEP 7 OF 19

z is not 0 and lastZone is 0: it was just pushed.
Go in.

VARIABLES

```
choice = 0
lastZone = 0
z = 1
```

SERIAL MONITOR

The menu, one pass at a time.

```
1  const char* items[] = {"Temperature", "Humidity", "Light"};
2  int choice = 0;
3  int lastZone = 0;
4
5  void loop() {
6      int z = zone(incipe.getJoystickXMovement());
7      if (z != 0 && lastZone == 0) {    // just pushed
8          choice = choice + z;
9          if (choice > 2) choice = 0;      // past the end: back to the start
10         if (choice < 0) choice = 2;      // before the start: to the end
11         Serial.println(items[choice]);
12     }
13     lastZone = z;
14     delay(50);
15 }
```

LINE 8 • STEP 8 OF 19

Move one item: 0 + 1.

VARIABLES

choice = 1
lastZone = 0
z = 1

SERIAL MONITOR

The menu, one pass at a time.

```
1  const char* items[] = {"Temperature", "Humidity", "Light"};
2  int choice = 0;
3  int lastZone = 0;
4
5  void loop() {
6      int z = zone(incipe.getJoystickXMovement());
7      if (z != 0 && lastZone == 0) {    // just pushed
8          choice = choice + z;
9          if (choice > 2) choice = 0;    // past the end: back to the start
10         if (choice < 0) choice = 2;    // before the start: to the end
11         Serial.println(items[choice]);
12     }
13     lastZone = z;
14     delay(50);
15 }
```

LINE 11 • STEP 9 OF 19

Show the new item.

VARIABLES

```
choice = 1
lastZone = 0
z = 1
```

SERIAL MONITOR

```
Humidity
```

The menu, one pass at a time.

```
1  const char* items[] = {"Temperature", "Humidity", "Light"};
2  int choice = 0;
3  int lastZone = 0;
4
5  void loop() {
6      int z = zone(incipe.getJoystickXMovement());
7      if (z != 0 && lastZone == 0) {    // just pushed
8          choice = choice + z;
9          if (choice > 2) choice = 0;      // past the end: back to the start
10         if (choice < 0) choice = 2;      // before the start: to the end
11         Serial.println(items[choice]);
12     }
13     lastZone = z;
14     delay(50);
15 }
```

LINE 13 • STEP 10 OF 19

Remember this pass: pushed.

VARIABLES

```
choice = 1
lastZone = 1
z = 1
```

SERIAL MONITOR

```
Humidity
```

The menu, one pass at a time.

```
1  const char* items[] = {"Temperature", "Humidity", "Light"};
2  int choice = 0;
3  int lastZone = 0;
4
5  void loop() {
6      int z = zone(incipe.getJoystickXMovement());
7      if (z != 0 && lastZone == 0) {    // just pushed
8          choice = choice + z;
9          if (choice > 2) choice = 0;    // past the end: back to the start
10         if (choice < 0) choice = 2;    // before the start: to the end
11         Serial.println(items[choice]);
12     }
13     lastZone = z;
14     delay(50);
15 }
```

LINE 6 • STEP 11 OF 19

Pass 3: you are still holding. Zone 1 again.

VARIABLES

```
choice = 1
lastZone = 1
z = 1
```

SERIAL MONITOR

```
Humidity
```

The menu, one pass at a time.

```
1  const char* items[] = {"Temperature", "Humidity", "Light"};
2  int choice = 0;
3  int lastZone = 0;
4
5  void loop() {
6      int z = zone(incipe.getJoystickXMovement());
7      if (z != 0 && lastZone == 0) {    // just pushed
8          choice = choice + z;
9          if (choice > 2) choice = 0;      // past the end: back to the start
10         if (choice < 0) choice = 2;      // before the start: to the end
11         Serial.println(items[choice]);
12     }
13     lastZone = z;
14     delay(50);
15 }
```

LINE 7 • STEP 12 OF 19

lastZone is 1 — this push was already counted.
Skip: one push, one move.

VARIABLES

```
choice = 1
lastZone = 1
z = 1
```

SERIAL MONITOR

```
Humidity
```

The menu, one pass at a time.

```
1  const char* items[] = {"Temperature", "Humidity", "Light"};
2  int choice = 0;
3  int lastZone = 0;
4
5  void loop() {
6      int z = zone(incipe.getJoystickXMovement());
7      if (z != 0 && lastZone == 0) {    // just pushed
8          choice = choice + z;
9          if (choice > 2) choice = 0;    // past the end: back to the start
10         if (choice < 0) choice = 2;    // before the start: to the end
11         Serial.println(items[choice]);
12     }
13     lastZone = z;
14     delay(50);
15 }
```

LINE 6 • STEP 13 OF 19

Later: you let go (lastZone becomes 0), then push the other way. Reading 10: zone -1.

VARIABLES

```
choice = 1
lastZone = 0
z = -1
```

SERIAL MONITOR

```
Humidity
```

The menu, one pass at a time.

```
1  const char* items[] = {"Temperature", "Humidity", "Light"};
2  int choice = 0;
3  int lastZone = 0;
4
5  void loop() {
6      int z = zone(incipe.getJoystickXMovement());
7      if (z != 0 && lastZone == 0) {    // just pushed
8          choice = choice + z;
9          if (choice > 2) choice = 0;      // past the end: back to the start
10         if (choice < 0) choice = 2;      // before the start: to the end
11         Serial.println(items[choice]);
12     }
13     lastZone = z;
14     delay(50);
15 }
```

LINE 8 • STEP 14 OF 19

Move back: $1 + (-1)$.

VARIABLES

choice = 0
lastZone = 0
z = -1

SERIAL MONITOR

Humidity

The menu, one pass at a time.

```
1  const char* items[] = {"Temperature", "Humidity", "Light"};
2  int choice = 0;
3  int lastZone = 0;
4
5  void loop() {
6      int z = zone(incipe.getJoystickXMovement());
7      if (z != 0 && lastZone == 0) {    // just pushed
8          choice = choice + z;
9          if (choice > 2) choice = 0;      // past the end: back to the start
10         if (choice < 0) choice = 2;      // before the start: to the end
11         Serial.println(items[choice]);
12     }
13     lastZone = z;
14     delay(50);
15 }
```

LINE 11 • STEP 15 OF 19

Show it.

VARIABLES

choice = 0
lastZone = 0
z = -1

SERIAL MONITOR

Humidity
Temperature

The menu, one pass at a time.

```
1  const char* items[] = {"Temperature", "Humidity", "Light"};
2  int choice = 0;
3  int lastZone = 0;
4
5  void loop() {
6      int z = zone(incipe.getJoystickXMovement());
7      if (z != 0 && lastZone == 0) {    // just pushed
8          choice = choice + z;
9          if (choice > 2) choice = 0;      // past the end: back to the start
10         if (choice < 0) choice = 2;      // before the start: to the end
11         Serial.println(items[choice]);
12     }
13     lastZone = z;
14     delay(50);
15 }
```

LINE 6 • STEP 16 OF 19

Let go, then push that way again: zone -1.

VARIABLES

choice = 0
lastZone = 0
z = -1

SERIAL MONITOR

Humidity
Temperature

The menu, one pass at a time.

```
1  const char* items[] = {"Temperature", "Humidity", "Light"};
2  int choice = 0;
3  int lastZone = 0;
4
5  void loop() {
6      int z = zone(incipe.getJoystickXMovement());
7      if (z != 0 && lastZone == 0) {    // just pushed
8          choice = choice + z;
9          if (choice > 2) choice = 0;      // past the end: back to the start
10         if (choice < 0) choice = 2;      // before the start: to the end
11         Serial.println(items[choice]);
12     }
13     lastZone = z;
14     delay(50);
15 }
```

LINE 8 • STEP 17 OF 19

$0 + (-1)$ is -1 — there is no box -1 .

VARIABLES

choice = -1

lastZone = 0

z = -1

SERIAL MONITOR

Humidity

Temperature

The menu, one pass at a time.

```
1  const char* items[] = {"Temperature", "Humidity", "Light"};
2  int choice = 0;
3  int lastZone = 0;
4
5  void loop() {
6      int z = zone(incipe.getJoystickXMovement());
7      if (z != 0 && lastZone == 0) {    // just pushed
8          choice = choice + z;
9          if (choice > 2) choice = 0;    // past the end: back to the start
10         if (choice < 0) choice = 2;    // before the start: to the end
11         Serial.println(items[choice]);
12     }
13     lastZone = z;
14     delay(50);
15 }
```

LINE 10 • STEP 18 OF 19

Before the start: wrap to the end.

VARIABLES

choice = 2
lastZone = 0
z = -1

SERIAL MONITOR

Humidity
Temperature

The menu, one pass at a time.

```
1  const char* items[] = {"Temperature", "Humidity", "Light"};
2  int choice = 0;
3  int lastZone = 0;
4
5  void loop() {
6      int z = zone(incipe.getJoystickXMovement());
7      if (z != 0 && lastZone == 0) {    // just pushed
8          choice = choice + z;
9          if (choice > 2) choice = 0;      // past the end: back to the start
10         if (choice < 0) choice = 2;      // before the start: to the end
11         Serial.println(items[choice]);
12     }
13     lastZone = z;
14     delay(50);
15 }
```

LINE 11 • STEP 19 OF 19

Show it.

VARIABLES

choice = 2
lastZone = 0
z = -1

SERIAL MONITOR

Humidity
Temperature
Light

The push button answers yes or no.

ONE LINE OF CODE

```
float pressed = incipe.getButtonResponse();
```

Button	You get
Pressed	1
Not pressed	0
Not detected	-1

Act when the button is released — the same edge idea as the stick. Wiki › Button.

Show the chosen reading on release.

```
1 float lastButton = 0;
2
3 void showChoice() {
4     switch (choice) {
5         case 0: Serial.println(incipe.getTemperature()); break;
6         case 1: Serial.println(incipe.getHumidity()); break;
7         case 2: Serial.println(incipe.getLightIntensity()); break;
8     }
9 }
10
11 // inside loop(), after lastZone = z;
12 float b = incipe.getButtonResponse();
13 if (lastButton == 1 && b == 0) showChoice(); // on release
14 lastButton = b;
```

LINE 1 • STEP 1 OF 9

Last pass's button reading: not pressed. The menu is on item 2, "Light".

VARIABLES

lastButton = 0

choice = 2

SERIAL MONITOR

Show the chosen reading on release.

```
1 float lastButton = 0;
2
3 void showChoice() {
4     switch (choice) {
5         case 0: Serial.println(incipe.getTemperature()); break;
6         case 1: Serial.println(incipe.getHumidity()); break;
7         case 2: Serial.println(incipe.getLightIntensity()); break;
8     }
9 }
10
11 // inside loop(), after lastZone = z;
12 float b = incipe.getButtonResponse();
13 if (lastButton == 1 && b == 0) showChoice(); // on release
14 lastButton = b;
```

LINE 12 • STEP 2 OF 9

You press: b is 1.

VARIABLES

lastButton = 0

choice = 2

b = 1

SERIAL MONITOR

Show the chosen reading on release.

```
1 float lastButton = 0;
2
3 void showChoice() {
4     switch (choice) {
5         case 0: Serial.println(incipe.getTemperature()); break;
6         case 1: Serial.println(incipe.getHumidity()); break;
7         case 2: Serial.println(incipe.getLightIntensity()); break;
8     }
9 }
10
11 // inside loop(), after lastZone = z;
12 float b = incipe.getButtonResponse();
13 if (lastButton == 1 && b == 0) showChoice(); // on release
14 lastButton = b;
```

LINE 13 • STEP 3 OF 9

Pressed last pass and released now?
lastButton is 0 — no.

VARIABLES

lastButton = 0

choice = 2

b = 1

SERIAL MONITOR

Show the chosen reading on release.

```
1 float lastButton = 0;
2
3 void showChoice() {
4     switch (choice) {
5         case 0: Serial.println(incipe.getTemperature()); break;
6         case 1: Serial.println(incipe.getHumidity()); break;
7         case 2: Serial.println(incipe.getLightIntensity()); break;
8     }
9 }
10
11 // inside loop(), after lastZone = z;
12 float b = incipe.getButtonResponse();
13 if (lastButton == 1 && b == 0) showChoice(); // on release
14 lastButton = b;
```

LINE 14 • STEP 4 OF 9

Remember: pressed.

VARIABLES

lastButton = 1

choice = 2

b = 1

SERIAL MONITOR

Show the chosen reading on release.

```
1 float lastButton = 0;
2
3 void showChoice() {
4     switch (choice) {
5         case 0: Serial.println(incipe.getTemperature()); break;
6         case 1: Serial.println(incipe.getHumidity()); break;
7         case 2: Serial.println(incipe.getLightIntensity()); break;
8     }
9 }
10
11 // inside loop(), after lastZone = z;
12 float b = incipe.getButtonResponse();
13 if (lastButton == 1 && b == 0) showChoice(); // on release
14 lastButton = b;
```

LINE 12 • STEP 5 OF 9

You let go: b is 0.

VARIABLES

lastButton = 1

choice = 2

b = 0

SERIAL MONITOR

Show the chosen reading on release.

```
1 float lastButton = 0;
2
3 void showChoice() {
4     switch (choice) {
5         case 0: Serial.println(incipe.getTemperature()); break;
6         case 1: Serial.println(incipe.getHumidity()); break;
7         case 2: Serial.println(incipe.getLightIntensity()); break;
8     }
9 }
10
11 // inside loop(), after lastZone = z;
12 float b = incipe.getButtonResponse();
13 if (lastButton == 1 && b == 0) showChoice(); // on release
14 lastButton = b;
```

LINE 13 • STEP 6 OF 9

Pressed last pass, released now — call showChoice().

VARIABLES

lastButton = 1

choice = 2

b = 0

SERIAL MONITOR

Show the chosen reading on release.

```
1 float lastButton = 0;
2
3 void showChoice() {
4     switch (choice) {
5         case 0: Serial.println(incipe.getTemperature()); break;
6         case 1: Serial.println(incipe.getHumidity()); break;
7         case 2: Serial.println(incipe.getLightIntensity()); break;
8     }
9 }
10
11 // inside loop(), after lastZone = z;
12 float b = incipe.getButtonResponse();
13 if (lastButton == 1 && b == 0) showChoice(); // on release
14 lastButton = b;
```

LINE 4 • STEP 7 OF 9

Which item? choice is 2.

VARIABLES

lastButton = 1

choice = 2

b = 0

SERIAL MONITOR

Show the chosen reading on release.

```
1 float lastButton = 0;
2
3 void showChoice() {
4     switch (choice) {
5         case 0: Serial.println(incipe.getTemperature()); break;
6         case 1: Serial.println(incipe.getHumidity()); break;
7         case 2: Serial.println(incipe.getLightIntensity()); break;
8     }
9 }
10
11 // inside loop(), after lastZone = z;
12 float b = incipe.getButtonResponse();
13 if (lastButton == 1 && b == 0) showChoice(); // on release
14 lastButton = b;
```

LINE 7 • STEP 8 OF 9

case 2: print the light reading. Say it is 700.

VARIABLES

lastButton = 1

choice = 2

b = 0

SERIAL MONITOR

700.00

Show the chosen reading on release.

```
1 float lastButton = 0;
2
3 void showChoice() {
4     switch (choice) {
5         case 0: Serial.println(incipe.getTemperature()); break;
6         case 1: Serial.println(incipe.getHumidity()); break;
7         case 2: Serial.println(incipe.getLightIntensity()); break;
8     }
9 }
10
11 // inside loop(), after lastZone = z;
12 float b = incipe.getButtonResponse();
13 if (lastButton == 1 && b == 0) showChoice(); // on release
14 lastButton = b;
```

LINE 14 • STEP 9 OF 9

Back in loop(): remember, released.

VARIABLES

lastButton = 0

choice = 2

b = 0

SERIAL MONITOR

700.00

Make the menu yours.

Step	Do this
1	Put your own limits into LOW_LIMIT and HIGH_LIMIT. If the menu moves the “wrong” way, swap what +1 and –1 do.
2	Add a fourth item, “Distance”, that prints <code>incipe.getDistance()</code> from the ultrasonic sensor (in cm). Find every line that has to change — there are four places.
3	Use the joystick’s own button instead of the push button. Plot <code>incipe.getJoystickButtonResponse()</code> first to find which value means “pressed”.

Decoding a TV remote is coming.

The IR receiver page on the Wiki says what the module does, but its incipe.* functions are not published yet.

This part will be added once they are — Session 7 builds on it.

Checkpoint 1

With LOW_LIMIT 25 and HIGH_LIMIT 75, what does zone(90) return?

READ THE CODE

```
zone (90) ;
```

ANSWER

1

WHY

90 is not -1 and not below 25, but it is above 75.

The joystick is unplugged. What does zone(-1) return?

READ THE CODE

```
zone (-1);
```

ANSWER

0

WHY

The first if catches -1 before it can count as “below 25”, so a missing joystick never moves the menu.

Checkpoint 3

choice is 2 (“Light”). You push; the zone is 1. What prints?

READ THE CODE

```
choice = choice + z;  
if (choice > 2) choice = 0;  
if (choice < 0) choice = 2;  
Serial.println(items[choice]);
```

ANSWER

Temperature

WHY

2 + 1 is 3, past the end, so line 2 wraps choice to 0.

Checkpoint 4

You push and hold the stick for 10 passes. How many lines print?

READ THE CODE

```
if (z != 0 && lastZone == 0) {  
    ...  
}  
lastZone = z;
```

ANSWER

1

WHY

Only the first pass is an edge; on the other 9, lastZone is already 1.

Checkpoint 5

You hold the button for 3 passes, then let go. How many times does showChoice() run?

READ THE CODE

```
if (lastButton == 1 && b == 0)
    showChoice();
lastButton = b;
```

ANSWER

Once

WHY

Only on the pass where lastButton is 1 and b is 0.

Three things to take away.

1

The joystick gives raw numbers. Plot them, pick two limits, and sort every reading into a zone — with -1 counting as resting.

2

Act on the edge — just pushed, just released — so one push moves one item, however long you hold.

3

A menu is an array plus a choice number that wraps around at both ends; the button's release edge selects.

Next — Session 6: Data logging with SD card. Saving readings to a file instead of only printing them.